

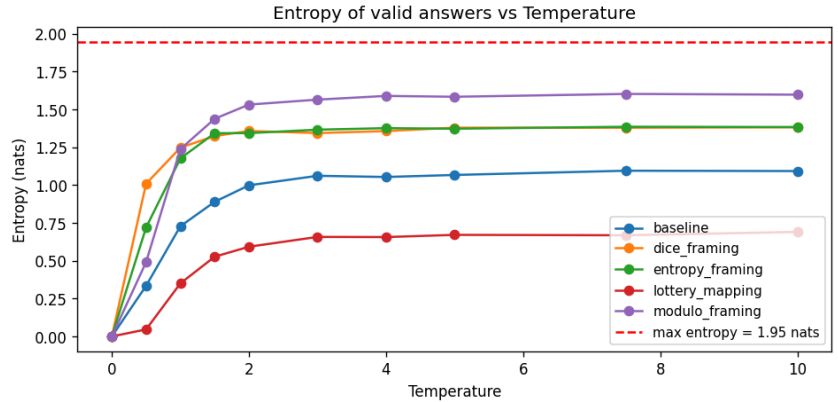
SDS 5350 Social Algorithm Assignment 3
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Part I: Single-Model Calibration [ALL CODEs ARE SAVED IN Part1_SingleModelCalibration.ipynb]

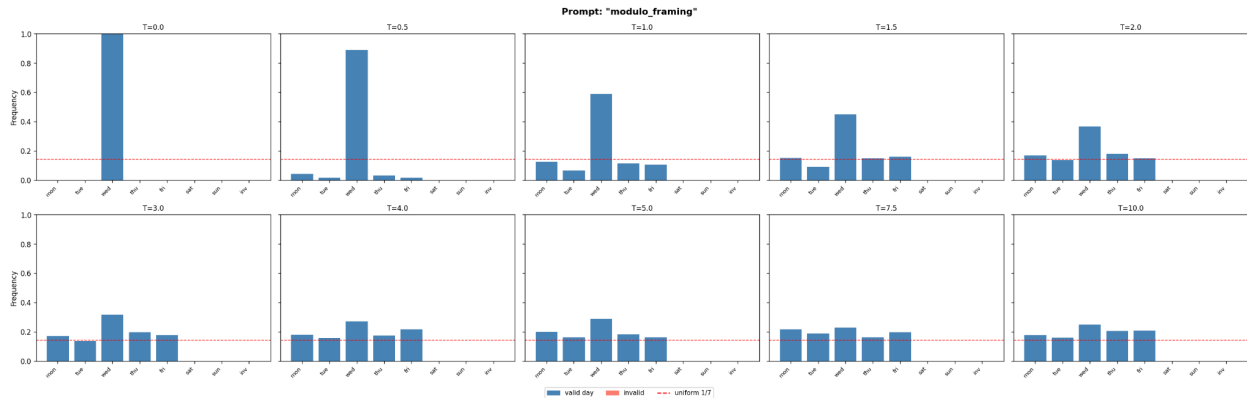
I.1: Warm-up: one simple Scattergories question

For this calibration, I used **qwen2.5:7b** through Ollama and sampled 500 outputs at each temperature in the grid. The target distribution is uniform over seven weekdays, so the maximum possible entropy is $\ln(7) \approx 1.946$ nats. I compared five prompt variants: baseline, dice_framing, entropy_framing, lottery_mapping, and modulo_framing.

- Baseline: Direct instruction to output exactly one lowercase weekday from the seven valid day names.
- Dice_framing: Role-play prompt that maps weekdays to faces of a fair 7-sided die and asks for one roll result.
- Entropy_framing: “Unbiased entropy generator” framing with an explicit seven-item list and strict equal-probability instructions.
- Lottery_mapping: Physical random-draw framing that maps a uniformly drawn number/slip to a weekday label.
- Modulo_framing: Algorithmic randomization prompt that samples a random integer, applies modulo 7, and maps the remainder to a weekday.



At temperature 0.0, all prompts were deterministic with entropy 0, although the deterministic day depended on prompt wording. baseline collapsed to sunday, while the other prompts mostly collapsed to wednesday. This confirms that prompt wording can change which answer dominates at low temperature, but cannot remove deterministic behavior.



The strongest prompt overall in the current run is **modulo_framing**(prompt detail in Appendix), which reached about 1.603 nats at T=7.5 and covered 5 weekdays (monday, tuesday, wednesday, thursday, friday). This is a clear improvement over the other prompt styles, but it is still below the uniform target of **1.946**. Coverage remains incomplete: across all these runs, Saturday never appeared, indicating persistent model bias even under high temperature and stronger randomization framing.

I.2 Next-token probabilities and top-k

For I.2, we queried Ollama directly for next-token logprob information using the same day-of-week setup (modulo framing) with top-k = 40. The returned top next-token distribution was highly skewed: “wed” had the largest mass (about 54.57%), followed by th (16.58%), fr (10.95%), then smaller probabilities on

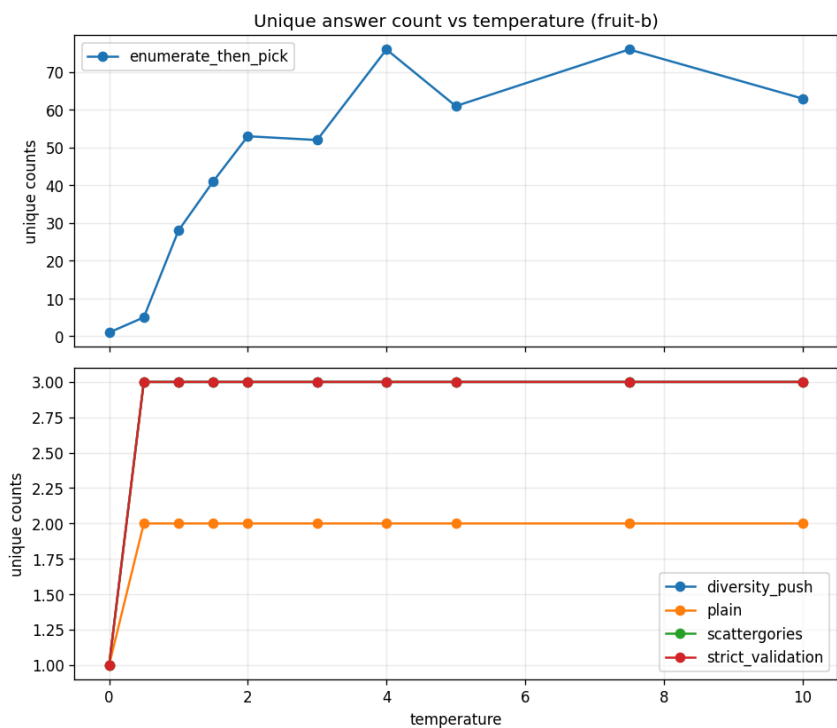
tokens like mon and s. These are subword tokens rather than full weekday strings, but they still provide useful directional evidence about bias in the model’s immediate next-token preferences. This pattern is consistent with the histogram results from I.1, where outputs were concentrated in a limited subset of weekdays and where “wednesday-like” continuations were strongly favored. In other words, the output-level imbalance in I.1 is already visible at the next-token level in I.2.

Token	Logprob	Probability (%)
wed'	-0.6056	54.57%
th'	-1.7969	16.58%
fr'	-2.2117	10.95%
t'	-2.466	8.49%
mon'	-3.0523	4.73%
s'	-3.083	4.58%
mond'	-8.0804	0.03%
friday'	-9.2348	0.01%
Wednesday'	-9.3525	0.01%
wed'	-9.463	0.01%

I.3 Harder prompt

In Part I.3, we repeated the same calibration pipeline on the harder prompt “fruits that start with b” and compared multiple prompt styles.

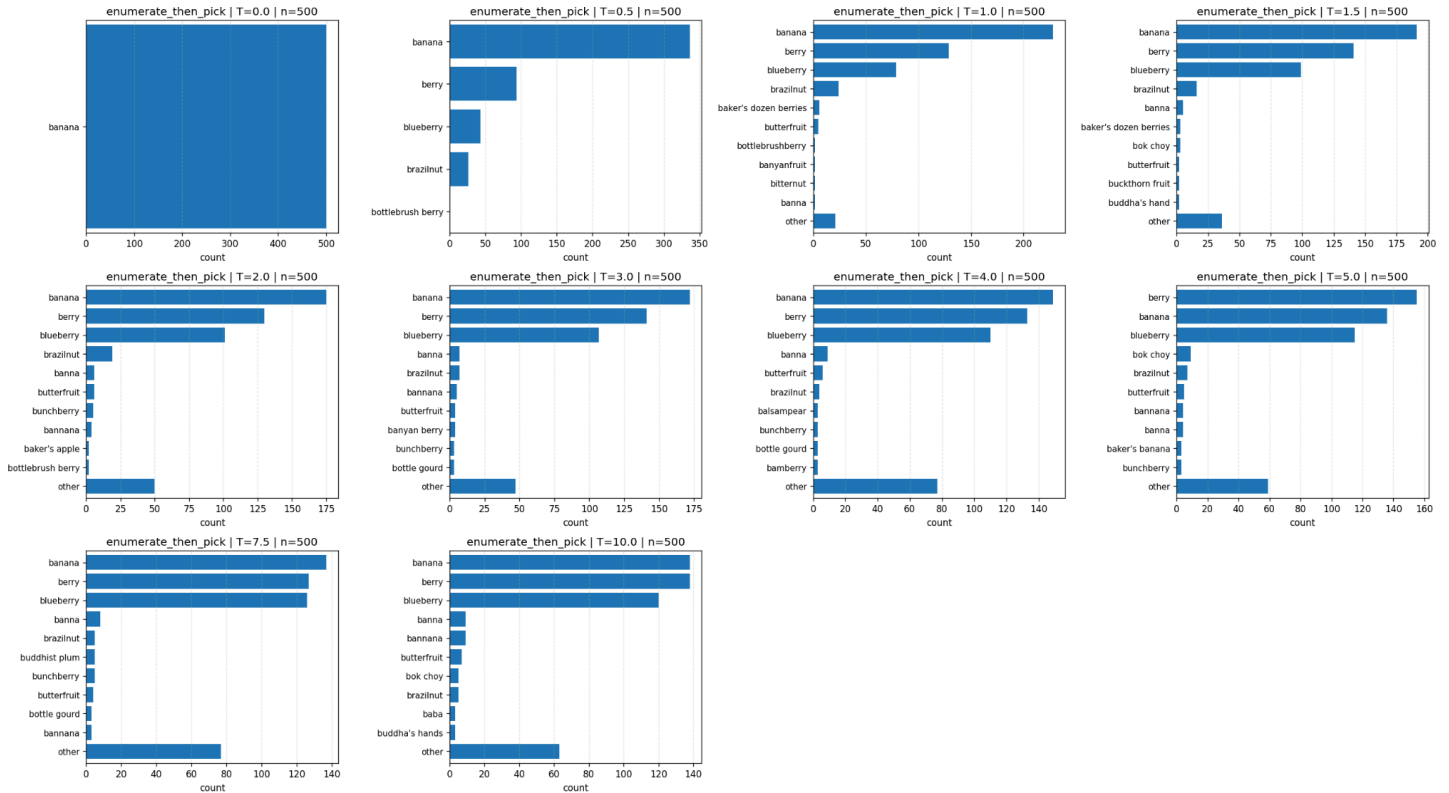
- Plain: A direct instruction to output exactly one fruit starting with b, with strict output formatting and no explanation.
- Scattergories: A game-framed version that explicitly says the model is playing Scattergories (letter=b, category=fruits) and should return one valid answer.
- Strict_validation: A validity-first prompt that adds explicit constraints (must be a real edible fruit, must start with b).
- Diversity_push: A prompt that encourages avoiding the most common answer while still requiring a valid b fruit and strict output format.
- Enumerate_then_pick: A strategy-oriented prompt that asks the model to internally think of multiple candidates and then output one final answer.



The most striking result is from **enumerate_then_pick** (prompt detail in Appendix). At low temperature, it is still deterministic (T=0.0: only banana, entropy 0). As temperature increases, its measured diversity grows very quickly (unique_answers reaches 76 at T=4.0 and T=7.5, with peak entropy about 2.281 at T=7.5). In contrast, the other prompt styles remain much narrower: plain stays at only 2 unique answers across temperatures, while scattergories, strict_validation, and diversity_push are mostly in the 2–3

unique-answer range. However, the diversity gain is partly artificial. Its outputs include many low-quality or invalid entries such as typos (banna, baker's dozen berries?), generic categories (berry), and non-fruit items (bok choy, bottle gourd). This is a key difference from the day-of-week task in I.1: day-of-week is a closed set with clear validity boundaries, while fruit-b is an open category where higher temperature and “enumerate then pick” prompting can inflate entropy by generating malformed or off-category answers.

prompt_id=enumerate_then_pick



Part II: Self-Play with One Model [ALL CODEs ARE SAVED IN Part2_SelfPlayWithOneModel.ipynb]

II.1 Game definition Phase A - generate answers

In Phase A, we used the same local model **qwen2.5:7b** twice as two separate players. Both players used the same prompt and temperature setting, and we generated two separate answer CSV files for self-play evaluation.

- prompt : scattergories_selfplay.txt
 - You are playing Scattergories.
 - Letter: {letter}
 - Category: {category}
 - Return exactly one valid Scattergories answer for this category that starts with the required letter.
 - Choose a common, realistic, category-appropriate answer.
 - Avoid explanations, lists, punctuation, and extra words.
 - Output only the answer in lowercase letters and spaces.
- temperature : 0.8
- rounds : 20

II.1 Game definition Phase B - judge and score

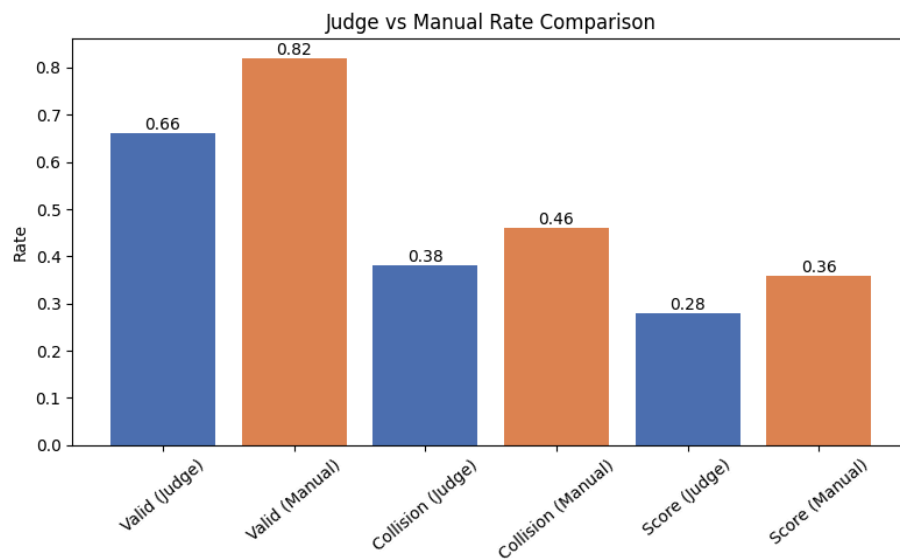
Judge Part Note : The assignment instruction indicates using gpt-5-mini with temperature=0 for deterministic evaluation. However, we encountered a technical issue where gpt-5-mini does not support temperature=0 (returning an unsupported value error). Following TF guidance, we instead used gpt-4o-mini with temperature=0 to ensure deterministic evaluation.

We used a judge.py file for judging and scoring. First, we randomly sampled 50 round_keys and checked the judged examples for both player1 and player2 (100 rows in total). We then manually verified each case. Selected examples are shown in the table below. The full audited examples and manually verified results are provided in audit_sample.csv.

player_id	round_key	letter	category	answer_norm	valid	collision	score	valid_manual	collision_manual	score_manual
answers_player1	Q001::15	A	Animals	albatross	1	1	0	1	1	0
answers_player2	Q001::15	A	Animals	albatross	1	1	0	1	1	0
answers_player1	Q004::3	B	Things that are red	banana	0	0	0	0	0	0
answers_player2	Q004::3	B	Things that are red	banana	0	0	0	0	0	0
answers_player1	Q006::16	C	Board games	chess	1	1	0	1	1	0
answers_player2	Q006::16	C	Board games	chess	1	1	0	1	1	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

The estimated judge error rates are shown in the table below. We observed cases where valid answers were marked as invalid, as well as cases where invalid answers were marked as valid. Because the collision and score calculations are based on the validity judgment, errors in validity also affect the collision and score outcomes. As a result, both the collision error rate and score error rate are non-zero. However, the valid error rate, collision error rate, and score error rate are all not that large.

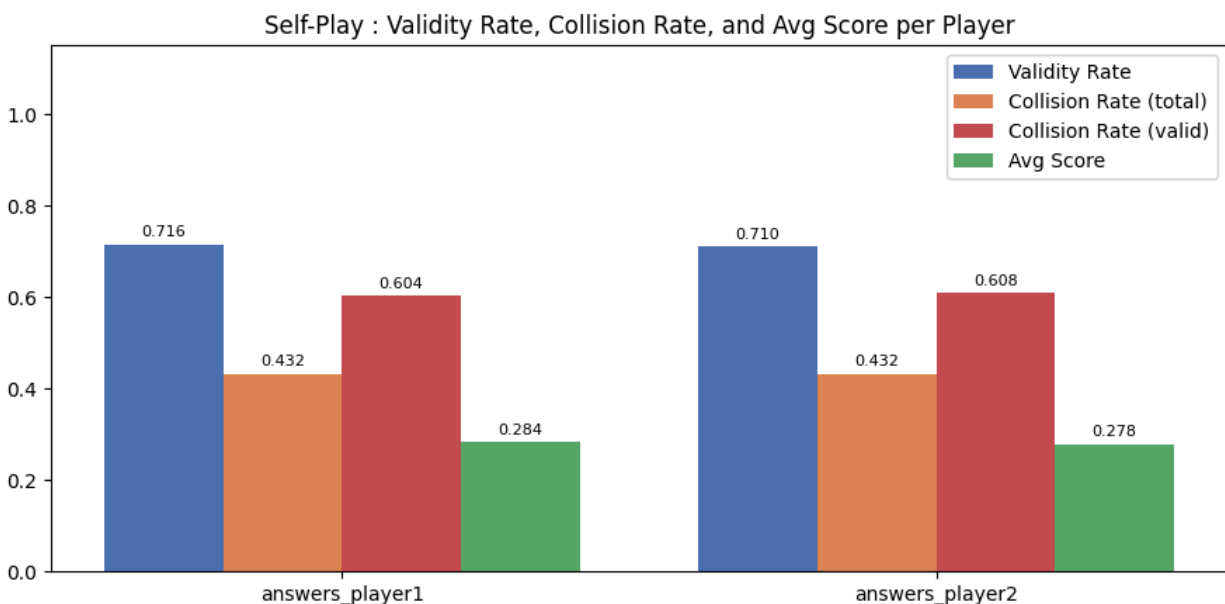
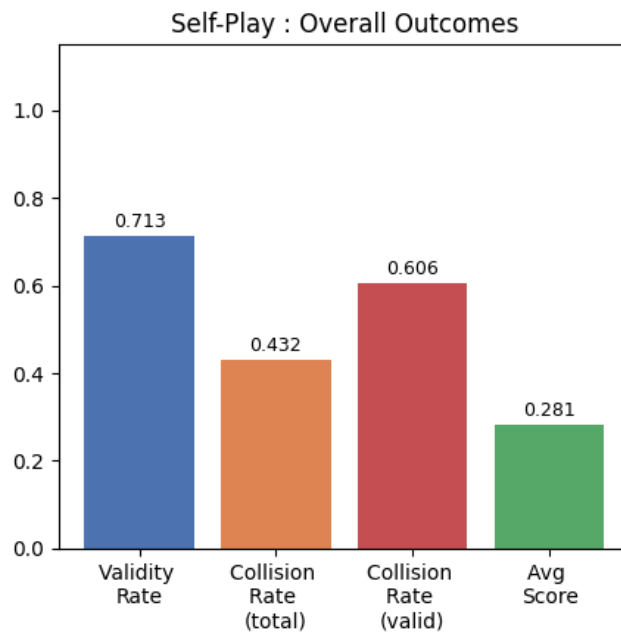
Valid Error Rate	Collision Error Rate	Score Error Rate
0.18	0.08	0.1



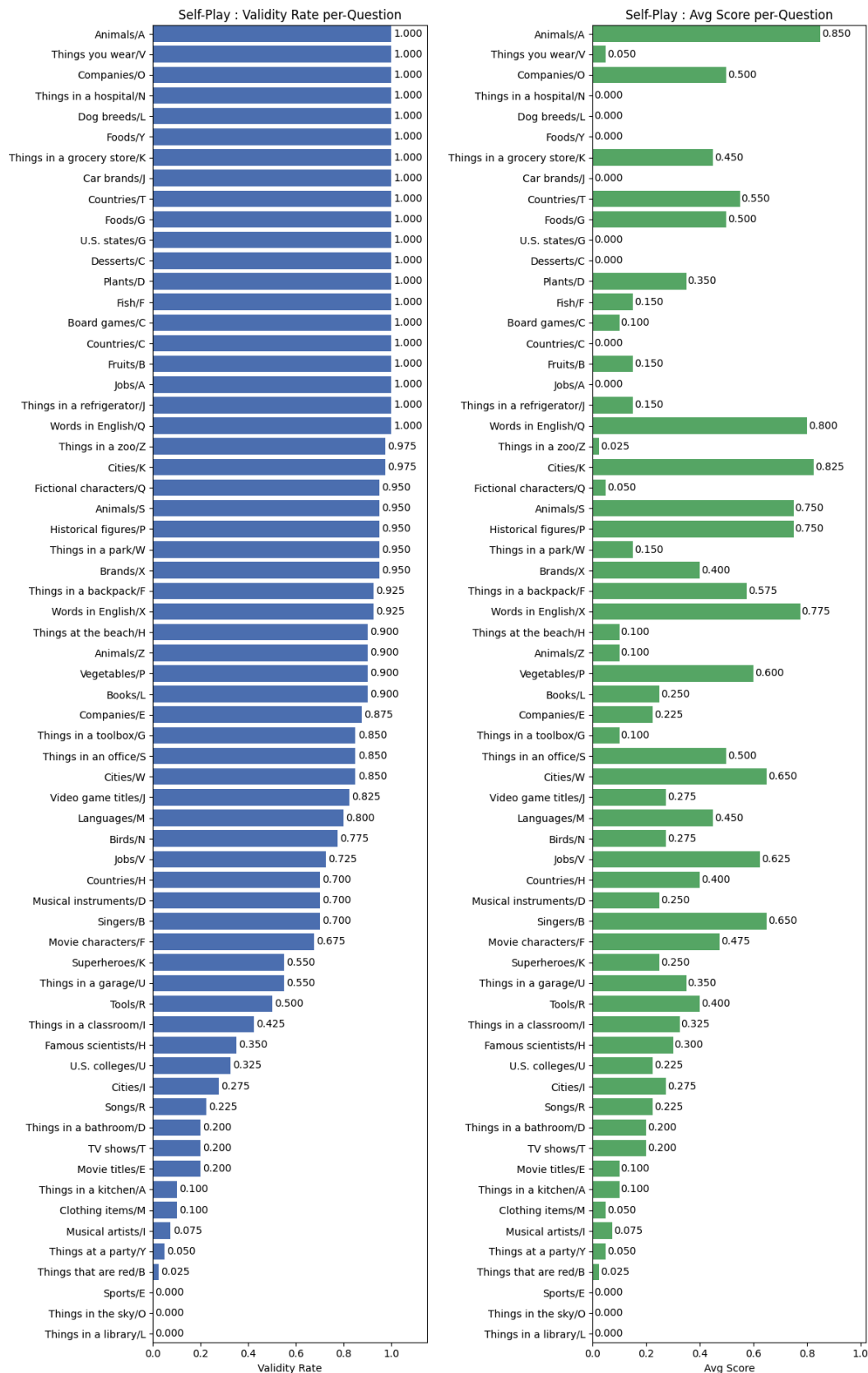
II.2 Self-play experiments

The overall outcomes of the self-play experiments are shown in the figure below. The average validity rate, defined as valid answers / total answers, is 0.713. The average collision rate, defined as collisions / total answers, is 0.432. The collision rate among valid answers, defined as collisions / valid answers, is 0.606. The average score is 0.281. Because collisions occur only among valid answers, we additionally report the collision rate among valid answers in order to better understand how frequently valid responses fail to earn points due to generating the same answers. The low average score of 0.281 shows that a deterministic strategy, which consistently selects the highest-probability answer, may not perform well in self-play competitive settings. In self-play, both players are sampled from the same output distribution, leading to a high collision rate and lower scores. Since points are awarded only when an answer is both valid and “unique”, similarity in output distributions directly reduces score.

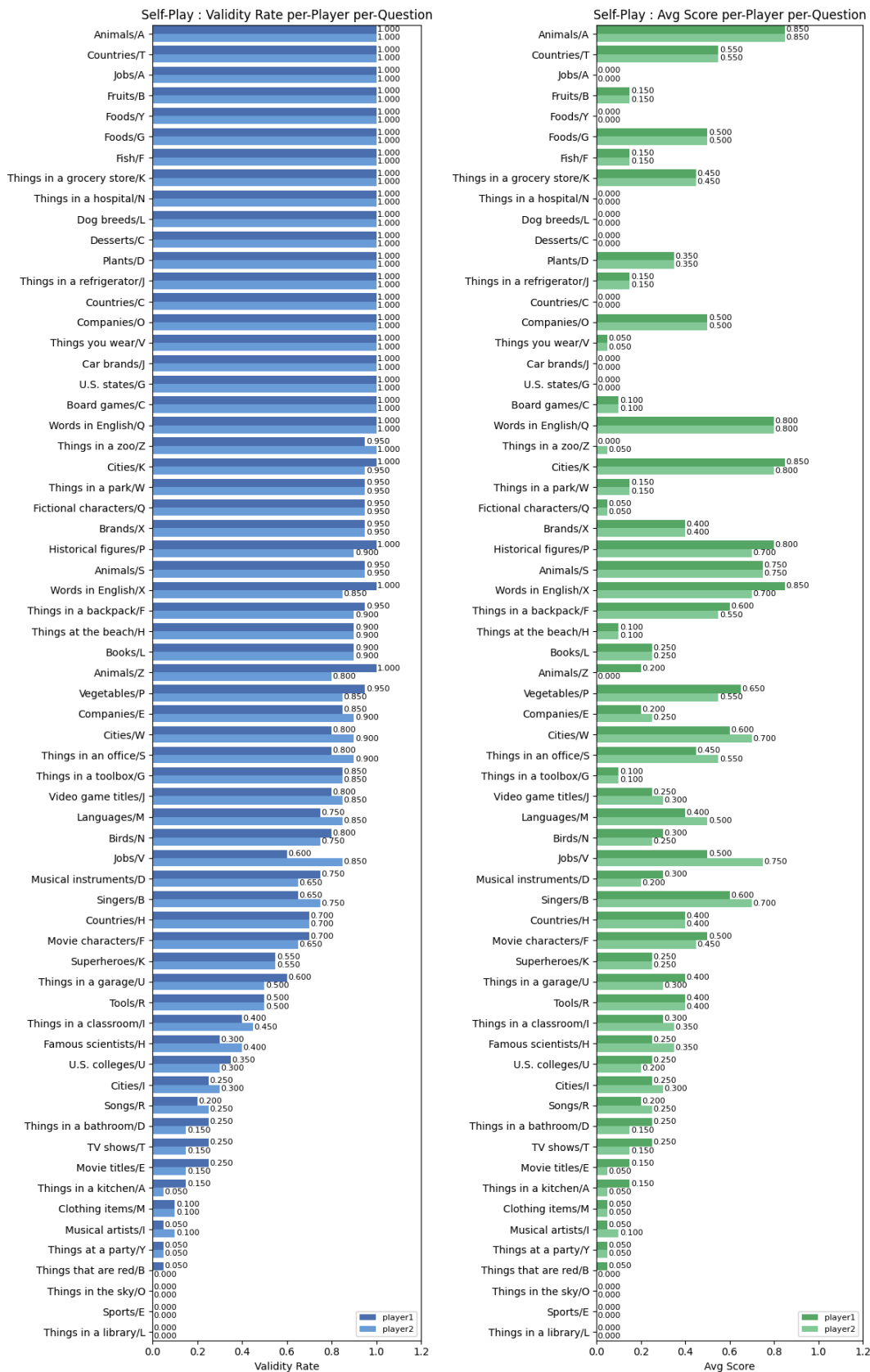
The per-player comparison plot shows that player 1 and player 2 achieved very similar results under identical settings.



The figures below show the per-question validity rate and average score.



The figures below show the per-question per-player validity rate and average score.

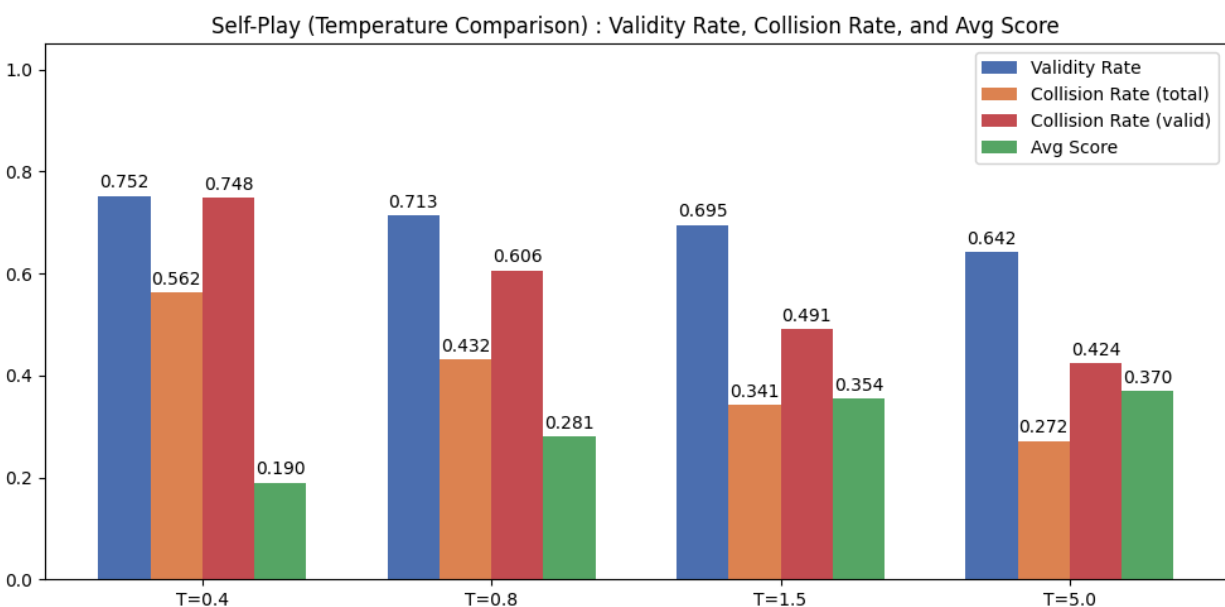


As shown in the figures above, there is huge variation in per-question validity rates and average scores. For categories that are broad and easy to answer (e.g., Animals/A), both the validity rate and the average score are relatively high. In contrast, for questions that are easy but have relatively few answers starting with the required letter (e.g., Things in a hospital/N), the validity rate is high but the average score is low due to frequent collisions. Since a point is given only when an answer is both valid and unique, questions with a lower validity rate show lower average scores.

The per-question, per-player validity rate and average score figures show that, since self-play uses the same model, the per-question validity rate and average score are similar across players.

To revisit the prompt and temperature choices and examine whether the self-play experiment score could be improved, we first varied the temperature setting while keeping the model, number of rounds, and prompt fixed. In addition to the baseline temperature of 0.8, we conducted self-play experiments at temperatures 0.4, 1.5, and 5.0. The figure below shows the validity rate, collision rate among total answers, collision rate among valid answers, and average score across different temperatures. We report only the overall outcomes of the self-play experiments across different temperatures because player 1 and player 2 have very similar scores under the same settings, as expected in self-play.

As temperature increases, the self-play experiment score improves. As discussed in Part I, higher temperature leads to a lower validity rate, since the model generates more diverse and less concentrated outputs. However, because in self-play experiments both players are generated by the same model and sampled from the same output distribution, lower temperatures result in more concentrated answers, increasing collisions and reducing scores. Thus, in self-play experiments, increasing temperature enhances output diversity, which reduces collisions and improves the average score, despite a decrease in the validity rate.



To further revisit the prompt and temperature choices and examine whether the self-play score could be improved, we tested an additional prompt while keeping the model, number of rounds, and temperature fixed. In addition to the baseline prompt (scattergories_selfplay.txt), we conducted a self-play experiment using an alternative prompt (scattergories_selfplay2.txt). Without making big changes to the overall structure of the prompt, we just added the phrase “avoid the most obvious default choices” in order to increase answer diversity and reduce collisions.

- prompt : scattergories_selfplay2.txt
 - You are playing Scattergories.
 - Letter: {letter}

Category: {category}

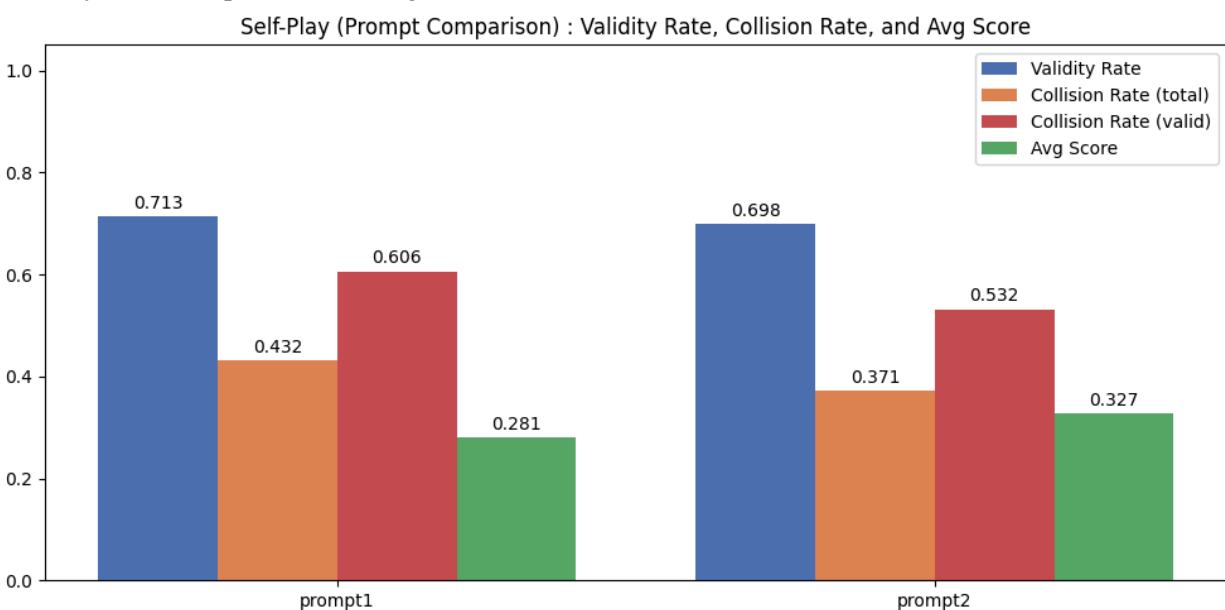
Return exactly one valid Scattergories answer for this category that starts with the required letter.

Choose a valid answer that is common, realistic, and appropriate for the category **but avoid the most obvious default choices.**

Do not include explanations, lists, punctuation, articles (a, an, the), and extra words.

Output only the answer in lowercase letters and spaces.

We report only the overall outcomes of the self-play experiments across different prompts because player 1 and player 2 have very similar scores under the same settings, as expected in self-play. After including the phrase “avoid the most obvious default choices”, the validity rate decreased slightly because we said to avoid “the most obvious default” choices. However, prompting the model to avoid the most obvious answers reduced collisions and resulted in a higher average score. This finding aligns with the earlier self-play experiments across different temperatures, which show that attempts to increase diversity and creativity tend to improve the average score.



Part III: Cross-Model Competition [ALL CODEs ARE SAVED IN Part3_CrossModelCompetition.ipynb]

III.1 Local model set

In Part 3, we used three local models **qwen2.5:7b**, **mistral:7b**, and **llama3.2:3b**. We conducted pairwise 2-player cross-model competitions for chosen local models. We used the same prompt and temperature settings across all models, and we generated one answer CSV file per player (model). We then judged each matchup (qwen2.5:7b vs mistral:7b, qwen2.5:7b vs llama3.2:3b, and mistral:7b vs llama3.2:3b) using the judge.py file. The baseline prompt, temperature, and number of rounds used in these experiments are shown below.

- prompt : scattergories_crossmodel.txt
 - You are playing Scattergories.
Letter: {letter}
Category: {category}
Return exactly one valid Scattergories answer for this category that starts with the required letter.
Choose a common, realistic, category-appropriate answer.
Avoid explanations, lists, punctuation, and extra words.

Output only the answer in lowercase letters and spaces.

- temperature : 0.8
- rounds : 20

III.2 Analysis

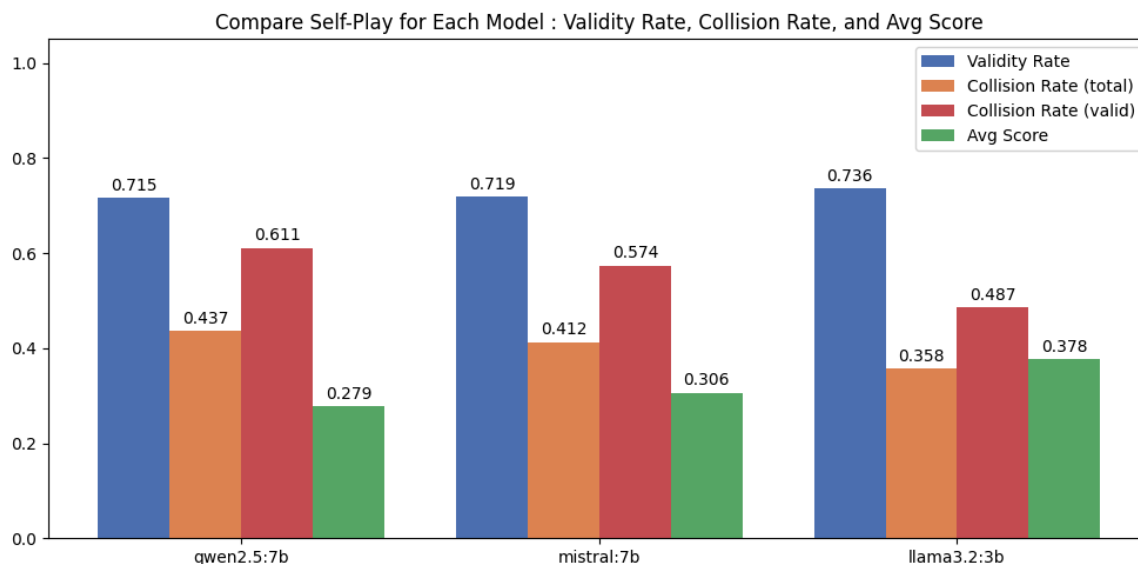
1) Self-play for each local model

The tables and figure below present the self-play experiment results for each model (qwen2.5:7b, mistral:7b, and llama3.2:3b). The first table reports the average self-play experiment results across both players for each model, while the second table reports the self-play experiment results separately for each player for each model. As expected in self-play, player 1 and player 2 have very similar scores under the same settings.

Regarding self-play experiments, llama3.2:3b showed the highest average score among the three models, due to lower collision rates and slightly higher validity rate. Qwen2.5:7b showed the highest collision rate among valid answers, which results in the lowest average score. This result shows that, in self-play experiments, even when validity rates are similar, differences in answer diversity across models can lead to differences in collision rates and average scores, as different models generate answers from different output distributions.

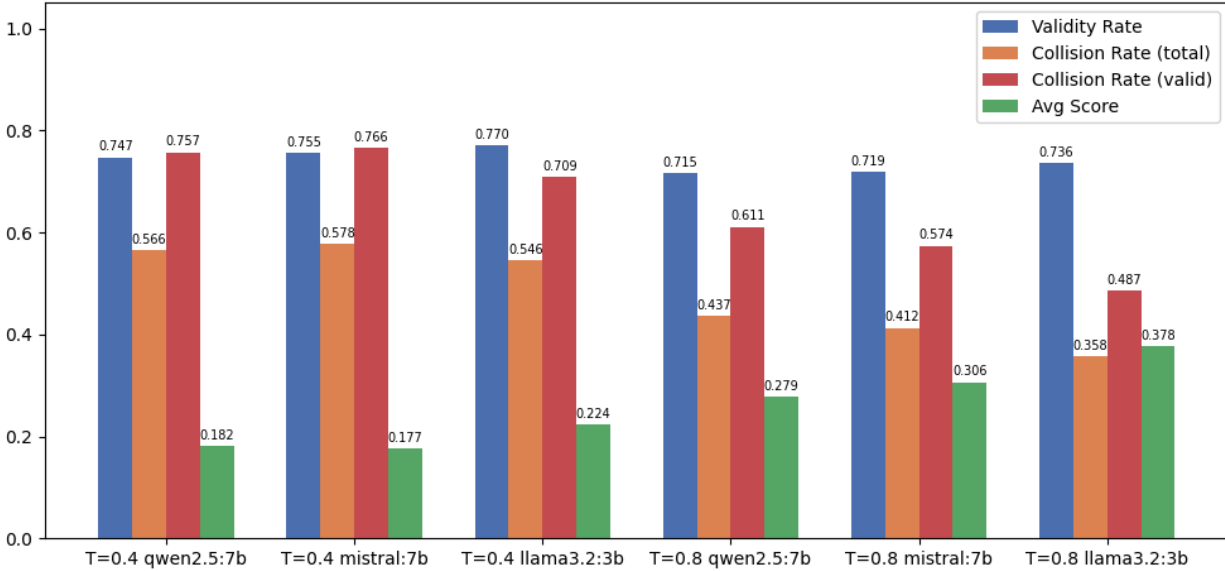
Model	Validity Rate	Collision Rate among Total Ans.	Collision Rate among Valid Ans.	Avg Score
qwen2.5:7b	0.715234	0.436719	0.610749	0.278516
mistral:7b	0.718750	0.412500	0.573914	0.306250
llama3.2:3b	0.735547	0.357812	0.486573	0.377734

Model		Validity Rate	Collision Rate among Total Ans.	Collision Rate among Valid Ans.	Avg Score
qwen2.5:7b	p1	0.726562	0.436719	0.601075	0.289844
	p2	0.703906	0.436719	0.620422	0.267188
mistral:7b	p1	0.717969	0.4125	0.574538	0.305469
	p2	0.719531	0.4125	0.573290	0.307031
llama3.2:3b	p1	0.746875	0.357812	0.479079	0.389062
	p2	0.724219	0.357812	0.494067	0.366406



We additionally examined how lowering the temperature affects self-play experiment results for each model. For all three models (qwen2.5:7b, mistral:7b, and llama3.2:3b), reducing the temperature to 0.4 increased the validity rate, consistent with the findings in Part I. However, due to the nature of self-play, where the same answers receive no points, the average score is lower at the low temperature ($T = 0.4$) compared to the baseline temperature ($T = 0.8$). It is because lower temperature concentrates the output distribution, increasing the likelihood that both players generate the same high-probability answers.

Compare Self-Play for Each Model (Temperature Comparison) : Validity Rate, Collision Rate, and Avg Score

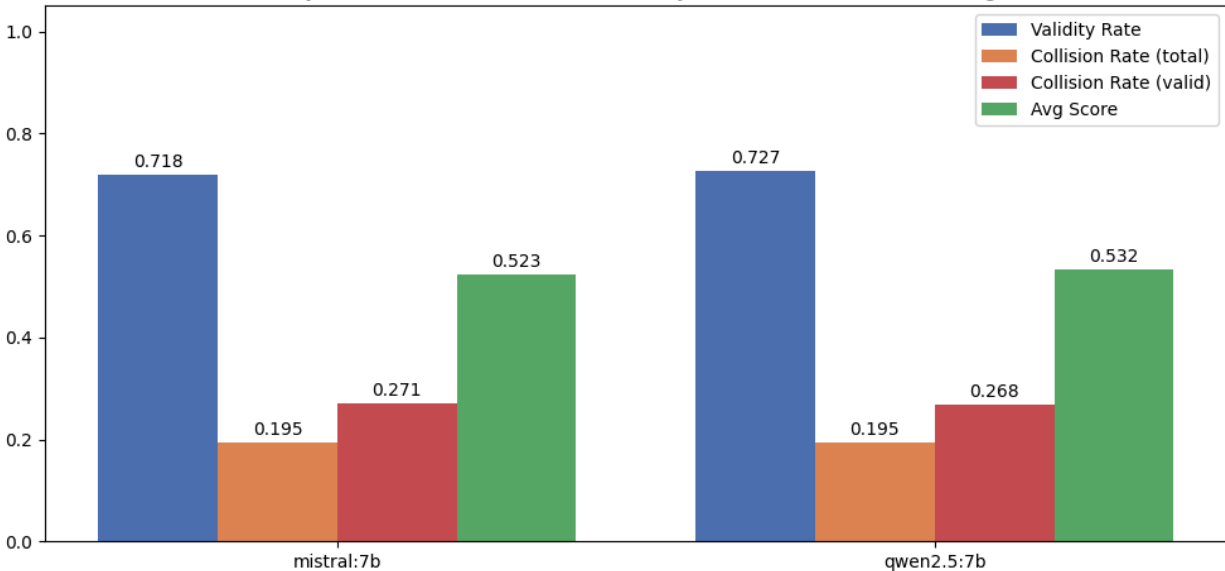


2) Cross-play for each pair of local models

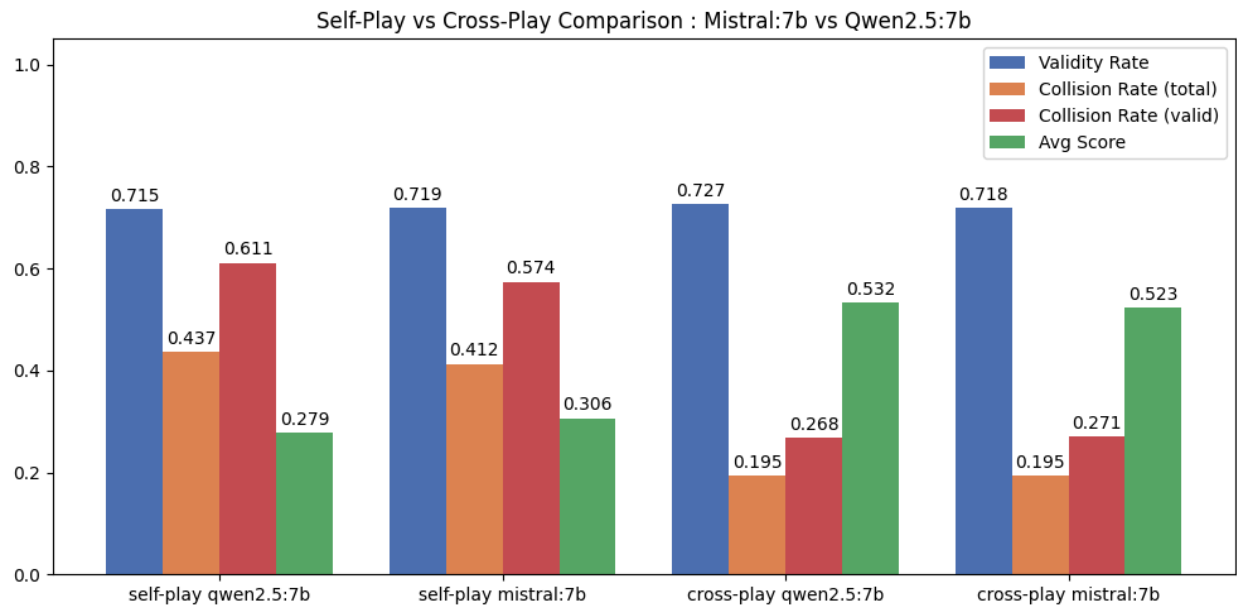
Qwen2.5:7b vs Mistral:7b

Model	Validity Rate	Collision Rate among Total Ans.	Collision Rate among Valid Ans.	Avg Score
qwen2.5:7b	0.726562	0.194531	0.267742	0.532031
mistral:7b	0.717969	0.194531	0.270947	0.523438

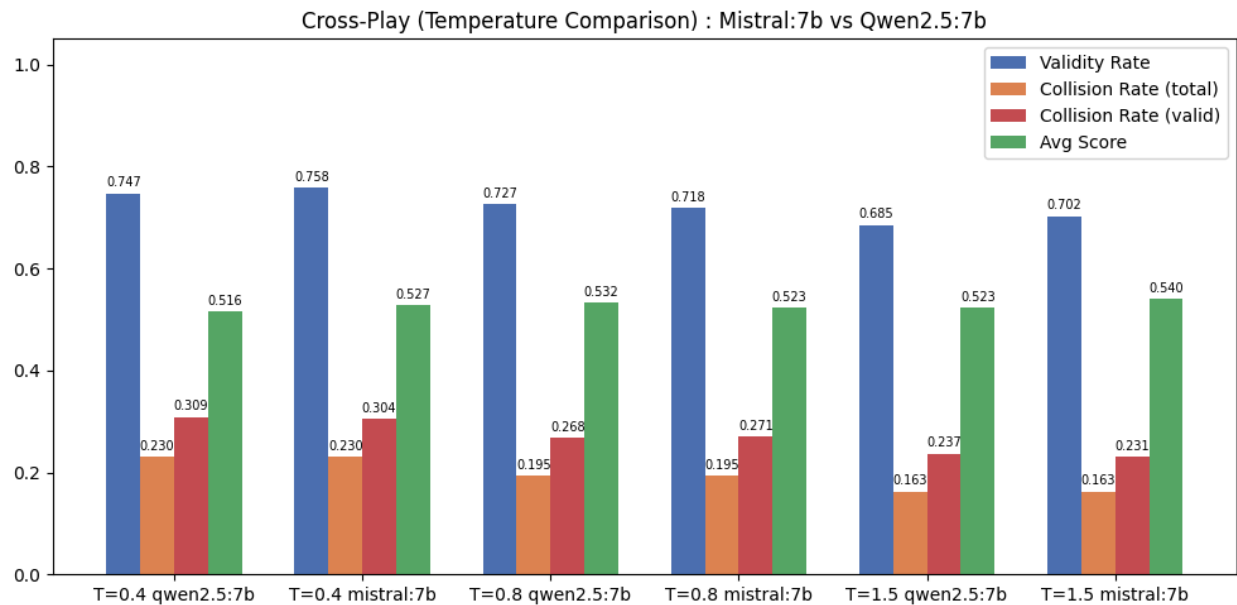
Cross-Play Mistral:7b vs Qwen2.5:7b : Validity Rate, Collision Rate, and Avg Score



The table and figure above show the results of the cross-play experiment between qwen2.5:7b and mistral:7b. In this matchup, qwen2.5:7b achieves a slightly higher average score than mistral:7b. Its validity rate is also slightly higher. However, validity rate, collision rate among total answers, collision rate among valid answers, and average score are very similar between the two models.

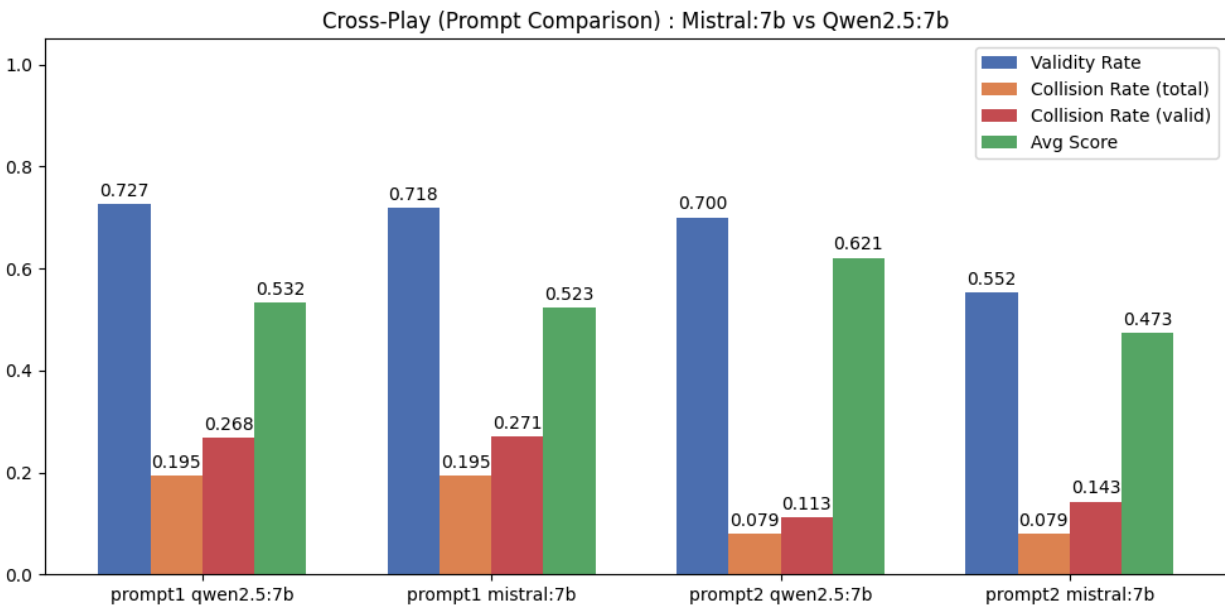


The figure above compares the self-play and cross-play results for qwen2.5:7b and mistral:7b. When comparing cross-play to self-play, validity rates remain very similar across settings. However, in the cross-play experiment, both models show substantially lower collision rates and higher average scores. This improvement happened because differences in output distributions across models reduce the likelihood of generating the same answers compared to self-play. Thus, both models achieve higher average scores in cross-play than in self-play.



The figure above explores variations in temperature in the cross-play experiment between qwen2.5:7b and mistral:7b. In addition to the baseline temperature of 0.8, we conducted cross-play experiments at a lower temperature (T=0.4) and a higher temperature (T=1.5). As temperature increases, both models show a

slight decrease in validity rates. At the same time, collision rates also decline as temperature increases. Higher temperature leads to more varied output answers, reducing overlap between the two models. In terms of average score, the changes are relatively small. For qwen2.5:7b, the average score increases slightly as temperature rises and then decreases slightly at the highest setting. For mistral:7b, the average score decreases slightly and then increases slightly. At T=0.8, qwen2.5:7b achieves a slightly higher average score than mistral:7b, whereas at T=0.4 and T=1.5, mistral:7b achieves a slightly higher average score.



The figure above explores variations in the prompt in the cross-play experiment between qwen2.5:7b and mistral:7b. For prompt 1, we used the file scattergories_crossmodel.txt. For prompt 2, we used the file scattergories_crossmodel2.txt, which includes the phrase “avoid the most obvious default choices” to limit overly obvious answers.

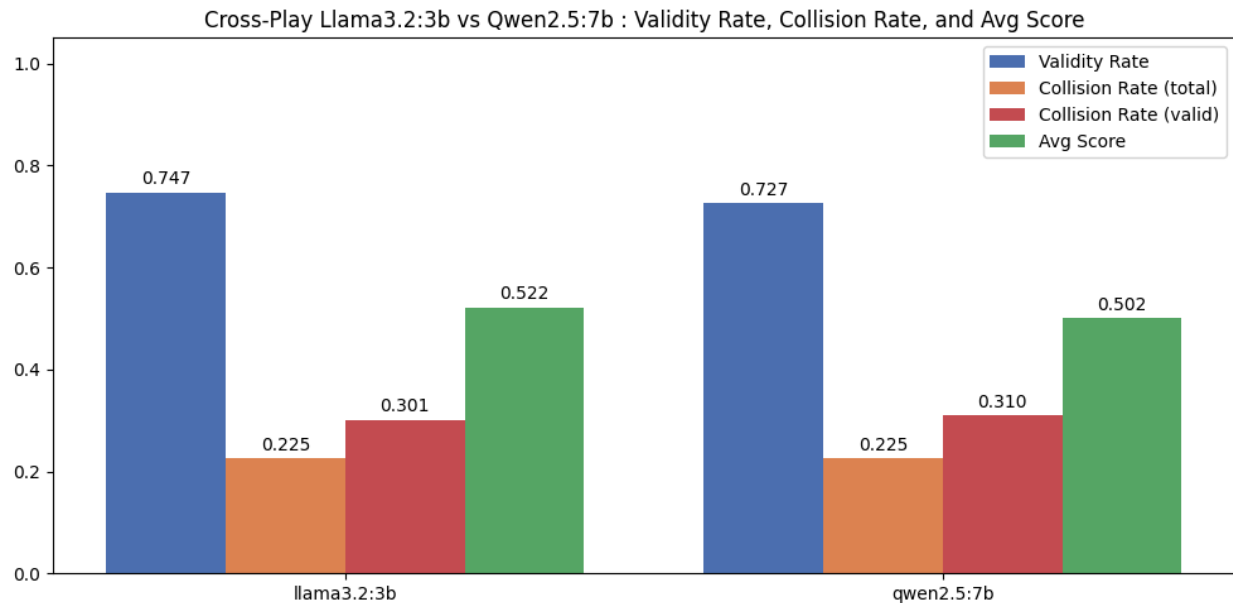
- prompt : scattergories_crossmodel2.txt
 - You are playing Scattergories.
 - Letter: {letter}
 - Category: {category}
 - Return exactly one valid Scattergories answer for this category that starts with the required letter.
 - Choose a valid answer that is common, realistic, and appropriate for the category **but avoid the most obvious default choices.**
 - Do not include** explanations, lists, punctuation, articles (a, an, the), and extra words.
 - Output only the answer in lowercase letters and spaces.

After adding the phrase “avoid the most obvious default choices,” the collision rate decreased by more than half for both models, showing that increased diversity reduced overlap in generated answers. However, there is an interesting difference between the two models. For qwen2.5:7b, adding this phrase did not affect the validity rate much. The model continued to generate valid answers while significantly reducing collisions, resulting in a much higher average score. In contrast, for mistral:7b, the same added phrase led to a significant decline in validity rate. While collisions decreased, the model struggled to generate valid answers under the prompt 2, which lowered its average score.

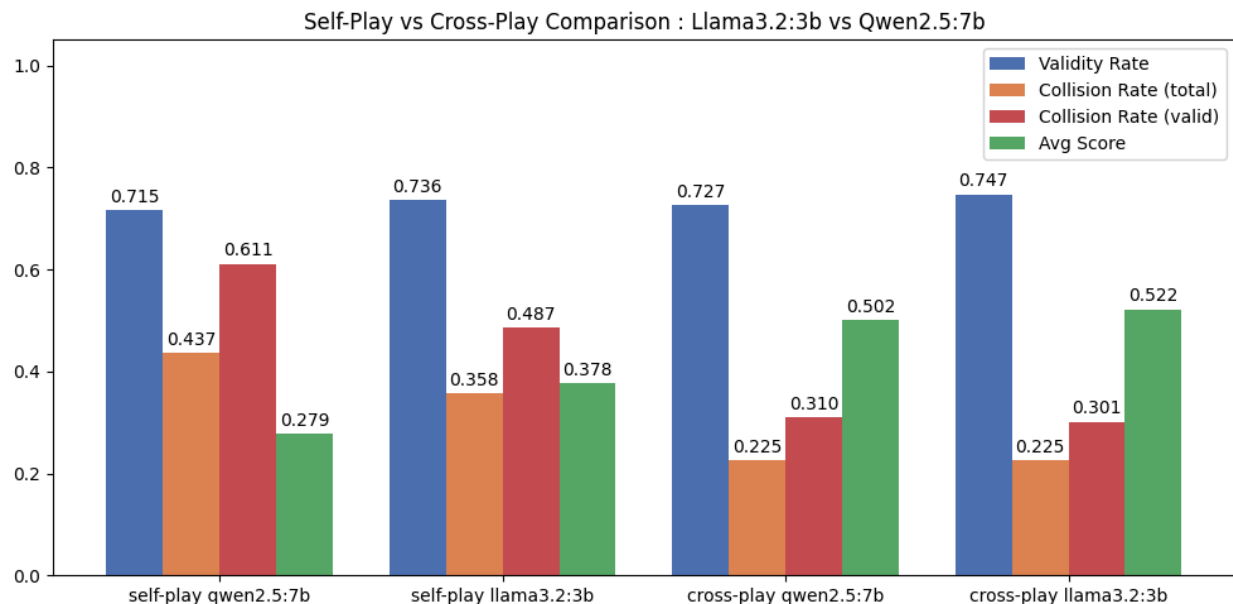
Qwen2.5:7b vs Llama3.2:3b

Model	Validity Rate	Collision Rate among Total Ans.	Collision Rate among Valid Ans.	Avg Score
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qwen2.5:7b	0.726562	0.225	0.309677	0.501563
llama3.2:3b	0.746875	0.225	0.301255	0.521875

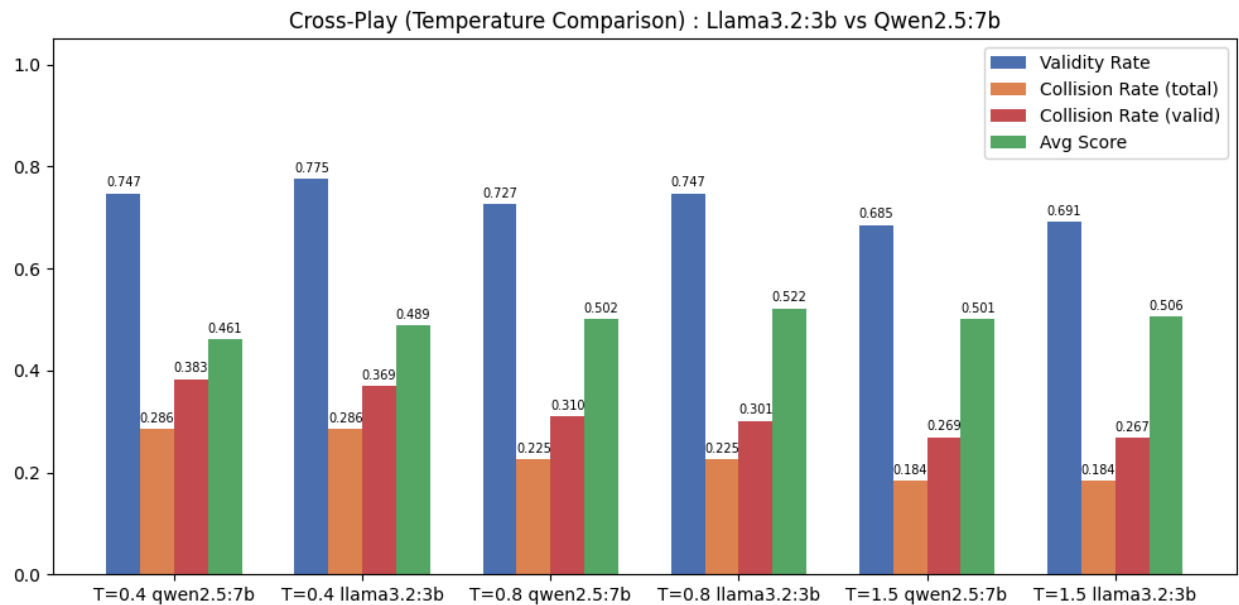


The table and figure above show the results of the cross-play experiment between qwen2.5:7b and llama3.2:3b. In this matchup, llama3.2:3b achieves a slightly higher average score than qwen2.5:7b. Its validity rate is also slightly higher. Although qwen2.5:7b (7B parameters) is larger than llama3.2:3b (3B parameters), the results show that a larger model size does not necessarily guarantee a higher validity rate or a competitive advantage in cross-play experiments.

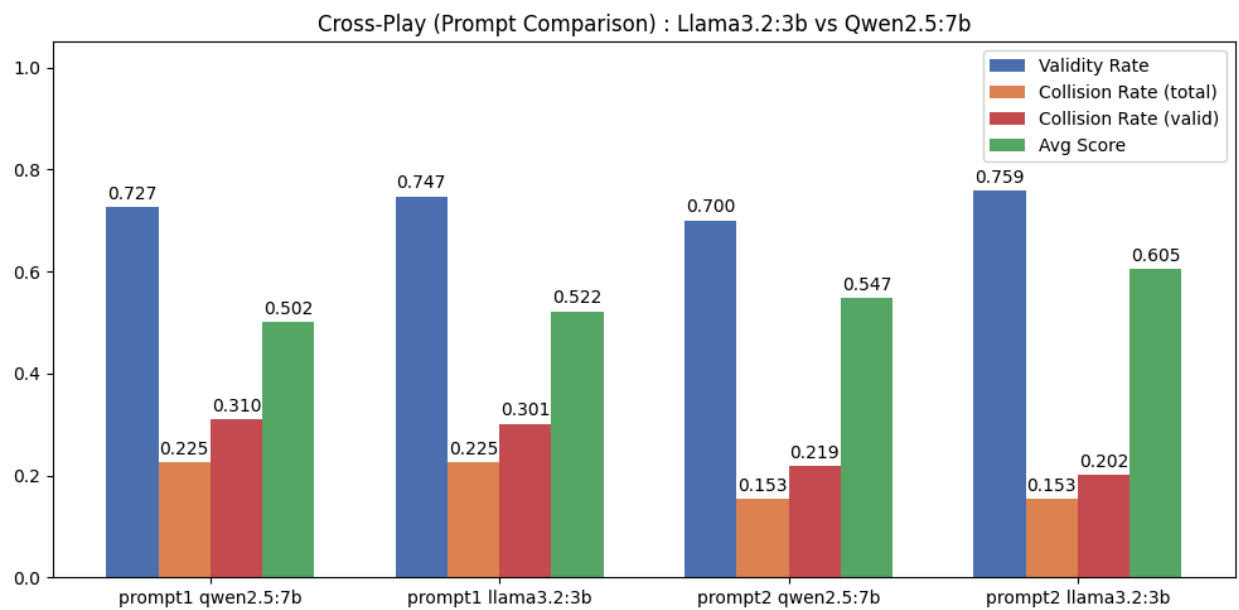


The figure above compares the self-play and cross-play results for qwen2.5:7b and llama3.2:3b. When comparing cross-play to self-play, validity rates remain very similar across settings for both models. However, in the cross-play experiment, both models exhibit substantially lower collision rates and noticeably higher average scores. This improvement occurs because differences in output distributions across models reduce the likelihood of generating overlapping answers compared to self-play. As a result, both qwen2.5:7b and llama3.2:3b achieve higher average scores in cross-play than in self-play. One

additional thing is that qwen2.5:7b showed a higher collision rate than llama3.2:3b in the self-play experiment, and experienced a larger reduction in collision rate when conducting cross-play. This shows that qwen2.5:7b is more vulnerable in self-play experiments.



The figure above explores variations in temperature in the cross-play experiment between qwen2.5:7b and llama3.2:3b. In addition to the baseline temperature of 0.8, we conducted cross-play experiments at a lower temperature (T=0.4) and a higher temperature (T=1.5). As temperature increases, both models show a slight decrease in validity rates. At the same time, collision rates also decline as temperature increases. Higher temperature leads to more varied output answers, reducing overlap between the two models. In terms of average score, the changes are small. For qwen2.5:7b, the average score increases slightly from T=0.4 to T=0.8 and then remains nearly unchanged at T=1.5. For llama3.2:3b, the average score increases slightly as temperature rises and then decreases slightly at the highest setting. At all three temperature settings (T=0.4, T=0.8, and T=1.5), llama3.2:3b achieves a slightly higher average score than qwen2.5:7b.

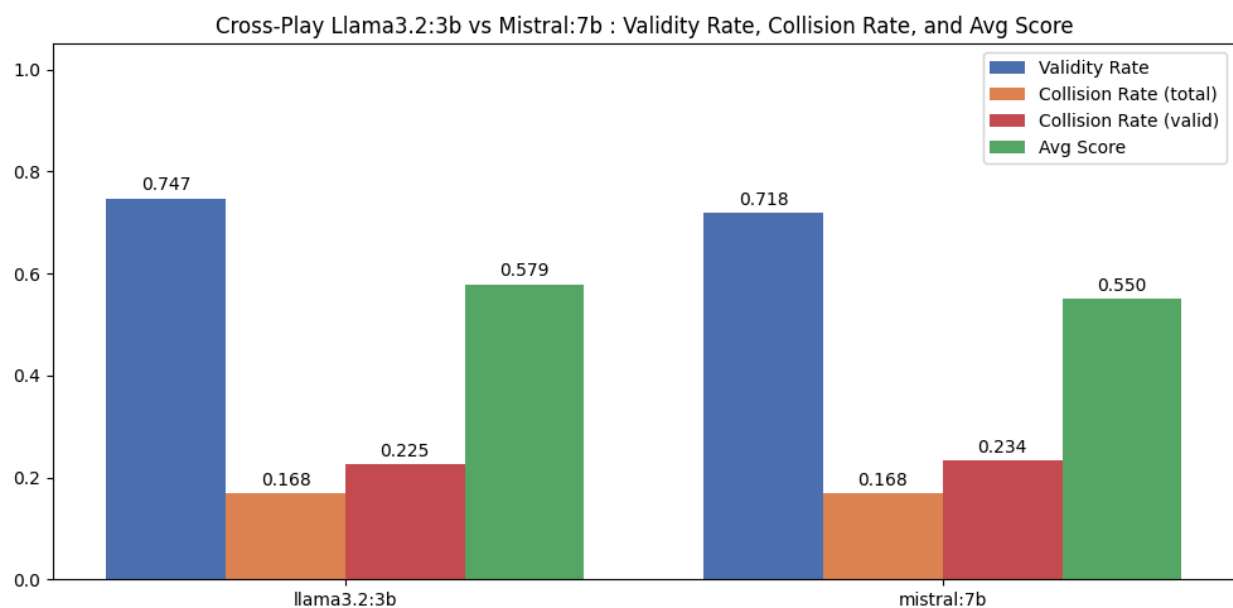


The figure above explores variations in the prompt in the cross-play experiment between qwen2.5:7b and llama3.2:3b. For prompt 1, we used the file scattergories_crossmodel.txt. For prompt 2, we used the file scattergories_crossmodel2.txt, which includes the phrase “avoid the most obvious default choices” to limit overly obvious answers.

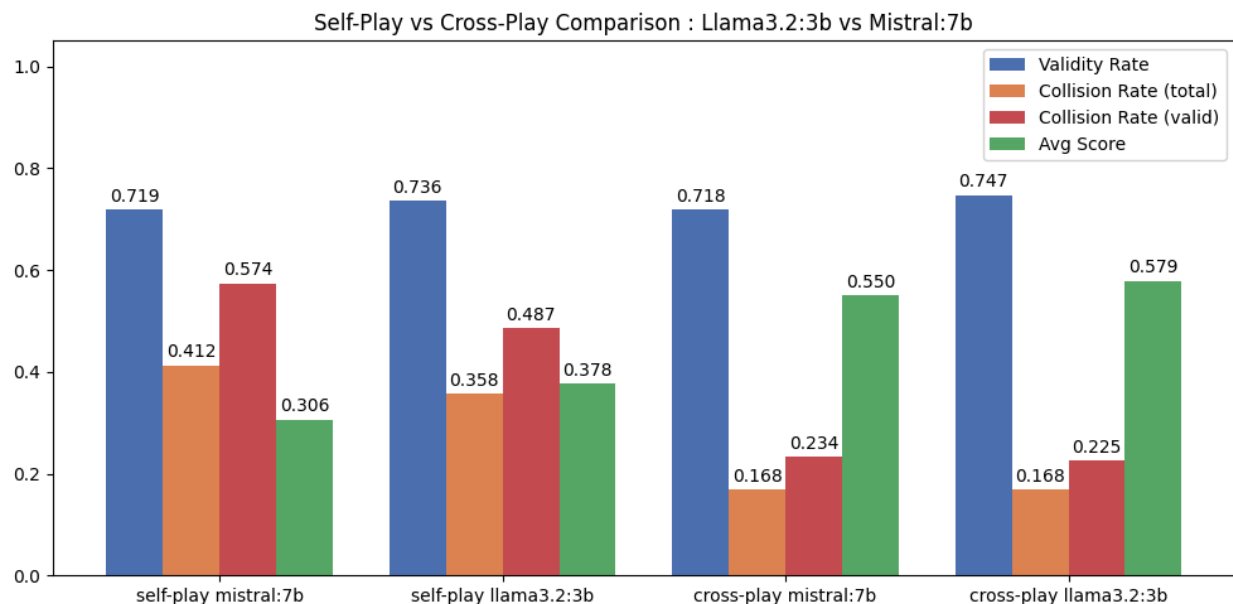
After adding the phrase “avoid the most obvious default choices”, the collision rate decreased for both models, showing that increased diversity reduced overlap in generated answers. Both models achieve higher average scores under prompt 2. However, there is an interesting difference. For qwen2.5:7b, adding this phrase slightly reduces the validity rate, indicating that limiting obvious default answers shows a modest trade-off between diversity and validity. Nevertheless, the reduction in collisions leads to an increase in average score. In contrast, for llama3.2:3b, validity rate increases very slightly even after adding the phrase “avoid the most obvious default choices”. As a result, llama3.2:3b showed bigger improvement in average score than qwen2.5:7b. Moreover, under both prompts, llama3.2:3b showed a higher average score than qwen2.5:7b.

Mistral:7b vs Llama3.2:3b

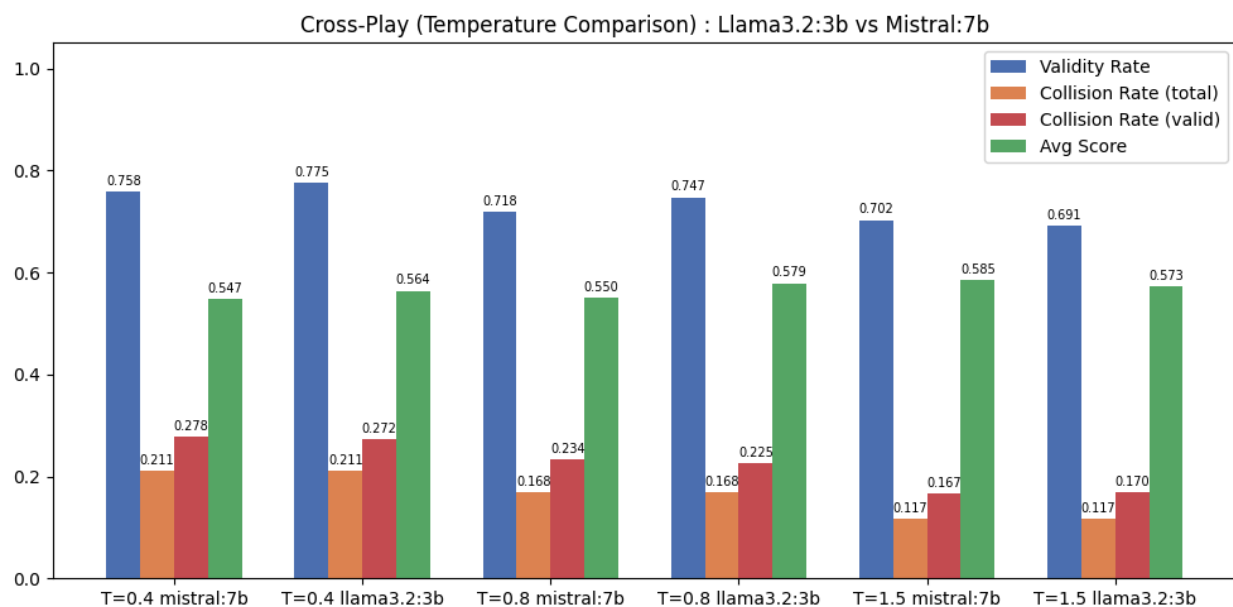
Model	Validity Rate	Collision Rate among Total Ans.	Collision Rate among Valid Ans.	Avg Score
mistral:7b	0.717969	0.167969	0.233950	0.550000
llama3.2:3b	0.746875	0.167969	0.224895	0.578906



The table and figure above show the results of the cross-play experiment between mistral:7b and llama3.2:3b. In this matchup, llama3.2:3b achieves a slightly higher average score than mistral:7b. Its validity rate is also slightly higher. Although mistral:7b (7B parameters) has more parameters than llama3.2:3b (3B parameters), the results show that a larger model size does not necessarily guarantee a higher validity rate or a competitive advantage in cross-play experiments.

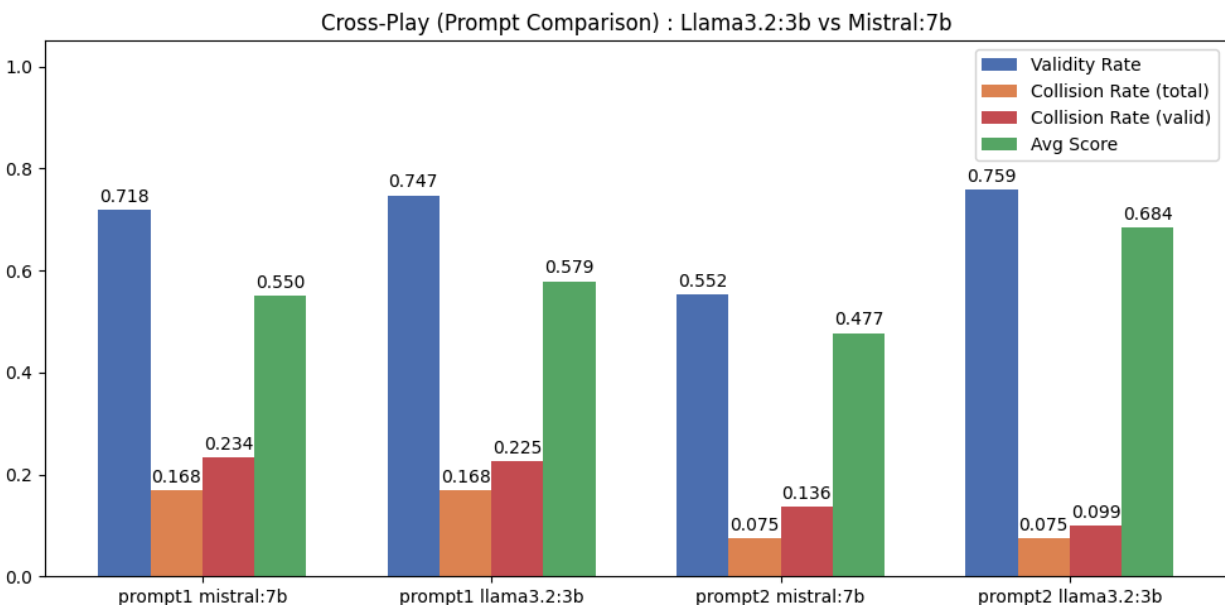


The figure above compares the self-play and cross-play results for mistral:7b and llama3.2:3b. When comparing cross-play to self-play, validity rates remain very similar across settings for both models. However, in the cross-play experiment, both models show significantly lower collision rates and higher average scores. This improvement occurs because differences in output distributions across models reduce the likelihood of generating the same answers compared to self-play. As a result, both mistral:7b and llama3.2:3b achieve higher average scores in cross-play than in self-play. Additionally, mistral:7b shows a higher collision rate than llama3.2:3b in self-play experiment, and experienced a larger reduction in collision rate when conducting cross-play. This shows that mistral:7b is more vulnerable in self-play experiments.



The figure above explores variations in temperature in the cross-play experiment between mistral:7b and llama3.2:3b. In addition to the baseline temperature of 0.8, we conducted cross-play experiments at a lower temperature (T=0.4) and a higher temperature (T=1.5). As temperature increases, both models show a slight decrease in validity rates. At the same time, collision rates also decline as temperature increases.

Higher temperature leads to more varied output answers, reducing overlap between the two models. In terms of average score, the changes are small. For mistral:7b, the average score increases slightly from T=0.4 to T=0.8 and increases again at T=1.5. For llama3.2:3b, the average score also increases from T=0.4 to T=0.8 and then slightly decreases at the highest temperature setting. At T=0.4 and T=0.8, llama3.2:3b has a slightly higher average score than mistral:7b, whereas at T=1.5, mistral:7b has slightly higher average score than llama3.2:3b. This shows that which model has a higher average score can change depending on the temperature setting.



The figure above explores variations in the prompt in the cross-play experiment between mistral:7b and llama3.2:3b. For prompt 1, we used the file scattergories_crossmodel.txt. For prompt 2, we used the file scattergories_crossmodel2.txt, which includes the phrase “avoid the most obvious default choices” to limit overly obvious answers.

After adding the phrase “avoid the most obvious default choices,” the collision rate decreased for both models, showing that increased diversity reduced overlap in generated answers. However, there is an interesting difference between the two models. For llama3.2:3b, adding this phrase did not affect the validity rate much. In fact, the validity rate increased slightly. The model continued to generate valid answers while significantly reducing collisions, resulting in a much higher average score. In contrast, for mistral:7b, similar to what we observed in the cross-play experiment with qwen2.5:7b, the same added phrase led to a big decline in validity rate. Although collisions decreased, the model struggled to generate valid answers while using prompt 2. Thus, the average score decreased under the prompt 2 setting. Under both prompts, llama3.2:3b showed a higher average score than mistral:7b.

In conclusion, when models do differently well against themselves than they do against other models, it shows that experiment and performance depends on the opponent. In self-play experiments, both players use the same model and thus generate answers from the same distribution, which leads to many collisions. If the temperature is low, the average score becomes even lower because the model tends to generate very similar or same answers repeatedly, further increasing collisions. In contrast, in cross-play experiments, different models generate answers from different distributions, so collisions decrease and average scores increase.

Both temperature and prompt play roles in these experiments. Low temperature makes the model more deterministic and increases the likelihood of generating the same answers, especially in self-play. Higher temperature generates more diverse answers, which reduces collisions and can improve the average score. The effect of temperature is more influential in self-play because both players already generate answers

from the same distribution, so lowering temperature increases overlap a lot. Similarly, adding a phrase that limits overly obvious answers can increase diversity and increase the average score. However, this often results in a slight decrease in validity rate, showing a trade-off between diversity and validity.

Reflection Questions:

1. For the day-of-week task, what prompt and sampling settings did you use, and how close did you get to uniform over valid answers? (Getting close to the uniform was surprisingly hard, I found, so it's OK if you can't get close to the uniform.)
For the day-of-week task, I used qwen2.5:7b in Ollama, sampled 500 generations per temperature, and compared five prompt variants: baseline, dice_framing, entropy_framing, lottery_mapping, and modulo_framing. The strongest setup was modulo_framing at $T=7.5$, which reached about 1.603 nats of entropy. Since the uniform target over seven weekdays is $\ln(7) \approx 1.946$ nats, this was still noticeably short of uniform. In practice, getting close to uniform was hard: coverage never reached all seven days, and Saturday never appeared in my runs, which suggests persistent model bias even under aggressive randomization framing and high temperature.
2. Were you able to extract next-token logprobs? If yes, how closely did they match empirical frequencies? If not, what blocked it?
Yes, I was able to extract next-token logprobs from Ollama. Using the day-of-week setup with top-k=40, the next-token distribution was already strongly skewed, with tokens corresponding to "wed" getting the largest probability mass, followed by prefixes for Thursday and Friday. These token-level probabilities did not match a uniform distribution, and they lined up directionally with the empirical sampling frequencies from Part I.1: where Wednesday dominated actual outputs.
3. For "fruits that start with b," which prompt variants did you test, and did explicitly framing it as Scattergories help or hurt?
For "fruits that start with b," I tested five prompt variants: plain, scattergories, strict_validation, diversity_push, and enumerate_then_pick. Explicitly framing the task as Scattergories did not help much in my runs. The scattergories prompt stayed in roughly the same narrow diversity range as plain and strict_validation, with only about 2 to 3 unique answers across temperatures, so it neither meaningfully improved diversity nor clearly improved exploration. The only prompt that substantially increased diversity was enumerate_then_pick, but that gain came with a majority of invalid or low-quality outputs, so the higher entropy there was partly artificial rather than a true improvement in useful answer diversity.
4. After revising prompt/temperature, how much did self-play performance change, and what do you think caused the change?
According to the part 2 self-play experiments, using the qwen2.5:7b model, revising the temperature and prompt led to an improvement in self-play performance. When temperature increased from 0.4 to 1.5 (even 5.0), the validity rate gradually decreased, but collision rates dropped a lot, which increased the average score from 0.190 ($T=0.4$) to 0.354 ($T=1.5$) and 0.370 ($T=5.0$). Increasing temperature from 0.4 to 1.5 increased the average score by 0.164 points. This suggests that in self-play, increasing diversity is better than maintaining a very high validity rate. Similarly, changing the prompt by adding "avoid the most obvious default choices" reduced collision rates among total answers from 0.432 to 0.371 and increased the average score from 0.281 to 0.327, even though validity decreased slightly. In conclusion, the improvement in self-play performance was mainly due to increased diversity in generating answers. Especially in self-play, both players generate answers from the same distribution, so overly obvious answers cause a lot of overlap. Reducing this overlap improves self-play performance.

5. How did self-play outcomes differ from cross-model outcomes, and what does that imply about diversity across models?

Self-play outcomes differ from cross-model outcomes. In self-play, there are relatively more collisions and lower average scores because both players use the same model and thus generate answers from the same distribution. Since they follow the same policy, they tend to generate similar high probability answers, which increases overlap. In contrast, cross-model competitions show lower collision rates and higher average scores because different models generate answers from different distributions. This implies that there is behavioral diversity across models. Each model follows a different policy and has a different output distribution. For example, mistral:7b is more sensitive to the phrase of the prompt that limits overly obvious answers and shows low validity rate under that prompt, while other models showed better performance in average score. Therefore, the result that cross-model performance is generally better than self-play performance indicates diversity in policies and output distributions across models.

6. What role did temperature play in self-play vs cross-play performance?

Temperature played a role in both self-play and cross-play by controlling answer diversity. It is more than just a sampling parameter. It controls entropy and the spread of the output distribution. Low temperature makes the model more deterministic and increases the likelihood of generating the same high probability answers. Temperature seems more influential in self-play experiments, where both players already generate answers from the same distribution. When temperature is low in self-play, entropy is low and the distribution becomes more concentrated, which leads to repeated answers and increase in collisions. As a result, average scores decrease. In contrast, higher temperature generates more diverse answers, reducing collisions and increasing average scores in self-play experiments. However, in cross-play experiments, the effect of temperature on performance is not that influential. Although validity rates gradually decrease as temperature increases, average scores do not change a lot. This is because in cross-play experiments, different models already generate answers from different distributions, so answers are already less likely to overlap, even at lower temperatures.

7. Overall, how much of your performance gain came from better prompts vs better sampling settings?

Based on my results, most of the performance gain appears to come from better sampling settings rather than from prompt changes alone. Prompt revisions helped mainly by improving formatting compliance and slightly tightening validity, but they did not fundamentally solve the core problem: the model still tended to converge on the same common answers for many categories. In my fruit-b experiments, for example, simply reframing the task as Scattergories did not substantially expand useful answer diversity compared with plain prompting.

By contrast, sampling settings had a larger effect because they directly changed the diversity-validity tradeoff. Very low temperature produced highly repetitive outputs, which is especially costly in self-play because both players often collide on the same answer. Raising temperature increased diversity and reduced some of that repetition, although too much temperature also risked more invalid answers.

8. What do these experiments suggest about AI systems interacting with other AI systems in the wild?

These experiments suggest that when AI systems interact with other AI systems in the wild, similarity across systems can lead to algorithmic monoculture. As we checked in the self-play experiments, when two systems generate answers from the same distribution, they repeatedly generate similar outputs, leading to high overlap and reduced diversity. In contrast, when different AI systems interact, each with different policies and output distributions, overlap decreases and

outputs become more diverse. Our cross-play experiments show that using different models results in more diverse, valid, and unique answers (low collision rate). This suggests that also in real-world, diversity across AI systems can help prevent homogenization and generate more novel and creative outputs. Ultimately, this diversity may also improve overall social welfare.

9. What follow-up experiment would you run next, and why?

We would like to run n-player cross-model competition experiments for follow-up experiments. In this assignment, we focused on pairwise 2-player competitions for chosen local models, but we want to extend the setting to 3-player or 4-player competitions, where multiple models compete simultaneously. In addition, to run n-player cross-model competition experiments, we would like to define alternative scoring rules beyond the current binary rule (1 point if an answer is valid and unique). For example, because avoiding overlap becomes harder as the number of players increases, we could use a scoring rule that gives more points when an answer does not overlap among many players. We would like to run this follow-up experiment because, in the wild, AI systems usually interact with many other AI systems at the same time rather than just competing in two-player settings. Thus, n-player cross-model competition experiments may better reflect real-world settings and more realistically capture how diversity and competition work in the wild.

APPENDIX

Part I.1 “Module - framing” prompt:

Task: output exactly one lowercase day name.

Use this exact procedure:

- 1. Sample a random integer n uniformly from 0 to 999.*
- 2. Compute $r = n \bmod 7$.*
- 3. Map r to a day:*
 - 0 -> monday*
 - 1 -> tuesday*
 - 2 -> wednesday*
 - 3 -> thursday*
 - 4 -> friday*
 - 5 -> saturday*
 - 6 -> sunday*
- 4. Output only the mapped day name.*

Rules:

- Do not output n .*
- Do not output r .*
- Do not output any explanation, punctuation, or extra text.*

Part I.3 “Enumerate_then_pick” prompt:

Before answering, silently recall at least 8 different real fruits that start with b — ranging from common to uncommon varieties. Do not output this list.

Then assign each recalled fruit a number starting from 1.

Pick a number at random from 1 to your count, trying to avoid always picking the same position.

Output only the fruit at that number.

Rules:

- Must be a real, edible fruit.*
 - Must start with the letter b.*
 - No brand names, no vegetables, no dishes.*
- Output format: one lowercase word or phrase, no punctuation, no explanation.*